Advanced Empirical Finance: Topics and Data Science

Neural networks and learning the SDF

Stefan Voigt

University of Copenhagen and Danish Finance Institute (DFI)

Spring 2027

Today

- Feed-forward neural networks
- Training: backpropagation and regularization
- Learning the SDF with neural networks

Feed-forward neural networks

TBD

Training: backpropagation and regularization

TBD

Learning the SDF with neural networks

TBD

Before the next lecture

- **Exercise 8:** Neural networks
- **Read:** Jensen, Kelly, Malamud & Pedersen (2026)